Cristae: The inner mitochondrial membrane gives rise to finger like projections called cristae. These cristae increase the inner surface area (fold in inner membrane) of the mitochondria to hold variety of enzymes.

Oxysomes: The inner mitochondrial membrane bear minute regularly spaced tennis racket shaped particles known as oxysomes (F_1 particle). They involve in ATP synthesis.

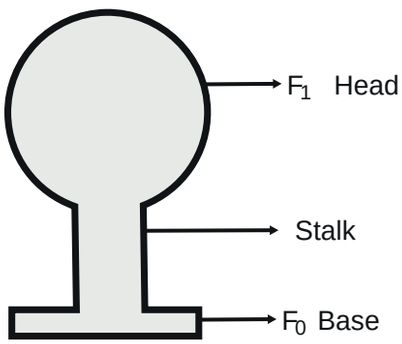


Figure 12.11 Structure of Oxysomes

Mitochondrial matrix - It is a complex mixture of proteins and lipids. Matrix contains enzymes for Krebs cycle, mitochondrial ribosomes(70 S), tRNAs and mitochondrial DNA.

12.10.2 Functions of Mitochondria

- Mitochondria is the main organelle of cell respiration. They produce a large number of ATP molecules. So they are called as **power houses of the cell** or **ATP factory of the cell**.
- It helps the cells to maintain normal concentration of calcium ions.
- It regulates the metabolic activity of the cell.

12.11 TYPES OF RESPIRATION

Respiration involves exchange of gases between the organism and the external environment. The plants obtain oxygen from their environment and release carbon dioxide and water vapour. This exchange of gases is known as **external respiration**. It is a physical process. Biochemical process occurs within cells where the food is oxidized to obtain energy, this is known as cellular respiration

12.11.1 Aerobic respiration

Aerobic respiration is the type of celluar respiration in which organic food is completely oxidized with the help of oxygen into carbon dioxide, water and energy. It occurs in most plants and animals.

$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + ATP$

Stages of Aerobic respiration

a. Glycolysis (Glucose splitting): It is the breakdown of one molecule of glucose (6 carbon) into two molecules of pyruvic acid (3 carbon). Glycolysis takes place in cytoplasm of the cell. It is the first step of both aerobic and anerobic respiration.

b. Krebs Cycle: This cycle occurs in mitochondria matrix. At the end of glycolysis, 2 molecules of pyruvic acid enter into mitochondria. The oxidation of pyruvic acid into CO_2 and water takes place through this cycle. It is also called **Tricarboxylic Acid Cycle** (TCA).

c. Electron Transport Chain: This is accomplished through a system of electron carrier complex called **electron transport chain** (ETC) located on the inner membrane of the mitochondria. $NADH_2$ and $FADH_2$ molecules formed during glycolysis and Krebs cycle are oxidised to NAD^+ and FAD^+ to release the energy via electrons. The electrons, as they move through the system, release energy which is trapped by ADP to synthesize ATP. This is called **oxidative phosphorylation**. In this

10th_Science_Unit-12.indd 183 06-12-2021 18:39:23